The Detector Was the Result: Eight Measurement Artifacts, and a Question That Cannot Be Asked This Way

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Abstract

Abstract. We report a simulation result, its retraction, and the audit that produced both. A protocell model of 1,845 Monte Carlo runs appeared to show that division regularity ("Clock") precedes spatial organization ("Map") without exception: 100 % consistency, binomial $p \approx 10^{-146}$. The result was reproducible, survived a twelve-condition parameter ablation, replicated across two geometries, and passed three rounds of external adversarial review. It was an artifact. The phase detector latched the second predicate only after the first had latched, so no other outcome was reachable; the reported inter-phase delay equalled the sampling interval exactly; and the first predicate's latch time tracked the moment its statistic became *computable* rather than any change in the system. We document eight such defects, each sufficient alone to produce the headline. We then validate the corrected measurement by planting a known ordering in the model's physics and recovering it ($\rho = +0.96$; Clock-first rising from 2.6 % to 100 % across the planted sweep), so that the corrected figure — the ordering holds in ≈1 % of runs — is a measurement rather than a blind spot. Three attempts to rescue the hypothesis follow. The third is the result: a Clock metric constructed specifically to have no definedness floor becomes defined *later* than the metric it replaces. A buffer sweep shows why. The replacement only measures regularity at all once its history spans ∼9 division periods ($\rho = -0.46$, $p = 0.003$ out of sample; shorter buffers return the wrong sign), and its definedness time equals that buffer length identically — so the earliest a valid Clock reading can exist is ∼24,000 ticks, against a spatial metric defined by ∼150. Measuring the regularity of a slow periodic process requires observing several of its periods; measuring correlation in a fast field does not. The asymmetry that produced the original artifact is therefore substantially a time–frequency constraint rather than an implementation error, and the ordering question is malformed as posed: comparing first-crossing times measures the ratio of two intrinsic timescales, not the dynamics. Ordering claims of this kind must be tested by intervention instead. We release an automated checklist that recovers all eight defects from source, and record that its authors committed the same class of error six times while writing this paper — the last of them an instrument validated by eye from an insignificant correlation, in the section arguing that unvalidated instruments are worthless.

Keywords: artificial life; protocell models; measurement artifact; phase detection; Simpson's paradox; reproducibility; self-correction

1. Introduction

1.1 A result that looked strong

In April 2026 we published a protocell simulation — in the tradition of minimal-cell and vesicle-replication modelling (Szostak et al., 2001; Hanczyc et al., 2003) and of the open problems programme in artificial life (Bedau et al., 2000) — reporting that the emergence of biological agency follows a mandatory order: temporal regularity of division (the *Clock*) precedes persistent spatial organization (the *Map*), which precedes thermodynamic selection (the *Engine*). Across 1,845 Monte Carlo runs in which both transitions occurred, the Clock preceded the Map in every single one.

By the conventional signals, the result was strong. Perfect separation. A binomial p-value of 8.01×10^{-146}. Replication across a one-dimensional ring and a two-dimensional spherical manifold. Robustness across a twelve-condition parameter ablation spanning factor-of-two perturbations in lipid supply, reaction–diffusion noise, growth perturbation, and stability window. Full public code and data from the day of posting. Three subsequent rounds of adversarial review by frontier language models, each of which found real problems and prompted genuine improvements, and none of which found what follows.

The result was an artifact. This paper is the account of how it was produced, how it was found, and what the finding turned out to be worth.

1.2 Thesis: reproducibility is not validity

Every number in the original paper reproduces. We re-ran the archived seeds during this audit and obtained identical phase-transition ticks a year later. The code does what it says. The data are real. Nothing was fabricated and nothing was tuned.

That is precisely the problem. A quantity can reproduce to fifteen significant figures and still be a number *of the wrong thing*. The reported inter-phase delay of 243 ticks is a correct arithmetic summary of a distribution whose median is 50 — and 50 was the sampling interval. The reported Clock latch time is a correct readout of when a statistic first returned a value — and that value was unavailable until a cell had divided four times.

We take this as the paper's thesis. Reproducibility protects against fabrication and against transcription error. It offers no protection at all against an instrument that cannot emit the answer you are looking for. The distinction is not new — computational reproducibility has long been argued to be necessary rather than sufficient (Peng, 2011; Stodden et al., 2016), and Leek and Peng

(2015) make the sharper point that reproducible research can still be wrong — but the case below is an unusually complete worked example, in which every number reproduces and the headline is nonetheless an artifact.

1.3 Contributions

1. Eight documented measurement defects in a published result, each independently sufficient to produce the headline, with the corrected figures.
2. A validated corrected measurement: we plant a known ordering in the physics and demonstrate the corrected method recovers it, so that our nulls are informative rather than blind.
3. The demonstration that the underlying question is malformed as posed — a time–frequency constraint, not a defect — and the constructive replacement.
4. `detector_audit.py`, an eight-check runner that recovers all eight defects from source and archived data with no audit knowledge encoded.
5. A record of six errors of the same class committed by the authors *during* the audit, which we take to be the most instructive part of the account.

1.4 Status of the original paper

The original preprint (SSRN 6593781, April 2026) remains posted with a correction notice attached rather than being removed. We took that decision deliberately. The paper's data and code are the evidence for this one, the defects are only legible against the original text, and silent withdrawal would leave the eleven months in which the claim circulated unaccounted for. The correction notice withdraws the central result explicitly and in full, lists all eight defects with corrected figures, and links to the audit materials. Readers arriving at the original by citation reach the correction before the abstract.

Two companion papers cannot be treated the same way. Neither the Vesicle Division nor the Engine study has surviving source code on any machine we control, so their headline figures — a division-interval CV = 0.06 and an effect size $g = 4.07$ — cannot be checked for the circularity documented here. We do not assert those results are wrong. We report that we are unable to verify them, which given what follows we consider the more useful statement.

2. The model and the original claim

2.1 Dynamics

The model follows a population of protocells. Each carries a Gray–Scott reaction–diffusion field (Gray and Scott, 1983; Pearson, 1993) on its membrane (a 24-node ring in 1D; a 642-vertex icosphere with a cotangent Laplace–Beltrami operator in 2D), grows by lipid accretion at a rate set by a shared resource pool, and divides by geometric instability once its reduced volume crosses a noisy threshold — the "Adder" mechanism (Taheri-Araghi et al., 2015; Campos et al., 2014). Pattern stability S couples multiplicatively into metabolic efficiency, nutrient uptake, and maintenance cost.

Division timing depends on membrane geometry and resource availability. It does not depend on the reaction–diffusion field. This one-way coupling was deliberate: it was intended to make the ordering claim non-circular.

2.2 The phase predicates

Three thresholded predicates define the phases:

- **Clock (B)**: population-mean division-interval coefficient of variation $\mathrm{CV} < 0.25$
- **Map (C)**: population-mean pattern stability $\bar{S} > 0.25$
- **Engine (D)**: $\bar{S} > 0.35$, $\mathrm{CV} < 0.3$, and generational depth ≥ 5

Each predicate, once satisfied, latches. The tick of first latching is recorded, and the ordering claim compares those ticks.

2.3 The published result

Table 1. The original headline, as published.

Quantity	Reported value
Clock before Map	1,845 / 1,845 (100 %)
Binomial p (1D)	8.01×10^{-146}
Binomial p (2D)	2.49×10^{-60}
Mean Clock→Map delay (1D)	$243 \pm 2{,}319$ ticks
Mean Clock→Map delay (2D)	56 ± 41 ticks
Hedges' g (1D)	9.41
Hedges' g (2D)	3.27

3. The audit

3.1 Method

The audit did not re-read the manuscript. It re-ran the simulation under added instrumentation, on the archived seeds, and compared what the instruments reported against what the dynamics did. Every defect below was found by running code against code, not by reasoning about the argument.

3.2 Defect 1 — the detector could not fail

The phase detector contains the following condition:

```
if ph.B and ph.B_tick < tick and not ph.C and mean_s > PHASE_C_S:
    ph.C = True
```

The Map predicate latches only if the Clock predicate latched at a *strictly earlier* tick. A run in which the Map preceded the Clock is not merely unlikely under this detector; it is unreachable. The reported 100 % was the only value the function could return.

Two consequences follow. First, the proportion carries no evidence about the dynamics. Second, both reported significance tests are computed against nulls that no possible run could have violated: the binomial tests $p = 0.5$ random ordering, and the Wilcoxon tests whether the delay exceeds zero, when the gate guarantees a delay of at least one sampling interval. A p-value against an unreachable null is not weak evidence. It is not evidence.

We re-instrumented both engines to log *ungated* first crossings: each predicate evaluated at every measurement tick, independently, with neither able to suppress the other. Default behaviour was unchanged (verified by exact reproduction of archived phase ticks on seeds 0, 1, and 2).

3.3 Defect 2 — the delay is the sampling interval

The reported mean delay was 243 ticks. The median is 50. The interquartile range is $[50, 50]$ — zero width. In 1D, 434 of 482 runs (90.0 %) recorded a delay of exactly 50 ticks; in 2D, 193 of 198 (97.5 %). Fifty ticks was the sampling interval.

Re-running 100 seeds at 1-tick sampling, the gated delay collapses to exactly 1 tick in 91 of the 97 runs in which both predicates latched (93.8 %; in the remaining three seeds the Map predicate never latched). The delay is not a property of the system. It is the measurement grid.

3.4 Defect 3 — the latch time is a definedness floor

The Clock statistic is undefined until a cell has accumulated three division intervals, which requires four completed divisions; until then it returns a sentinel value of 1.0, above every candidate threshold. The Clock predicate therefore *cannot* fire before that point, whatever the dynamics do.

Table 2. The reported Clock latch coincides with the tick at which the Clock statistic first becomes computable. Medians over the full archive ($n = 500$ in 1D, $n = 200$ in 2D).

	First tick CV computable	Reported Clock latch
1D	4,200	4,200
2D	4,900	4,900

An earlier draft reported 4,250/4,250 and 4,950/4,975 here. Those were medians over a 120-run subset; the figures above are the full archive. The agreement is exact in both geometries at full n. All such medians lie on a 50-tick grid, because 50 ticks is the sampling interval; a median quoted off that grid is an average of two adjacent grid points across an even number of runs, not an observation. An earlier draft reported 4,825 for the 1D figure on that basis.

Meanwhile the Map statistic exceeds its threshold at median tick 1,400 (1D) and 1,300 (2D) — before the Clock statistic exists at all. The Map predicate crossed before CV was computable in 115 of 116 archived 1D runs and 118 of 118 2D runs.

We initially catalogued this as a straightforward implementation error. Section 5.3 shows that assessment was only half correct.

3.5 Defect 4 — the mean is three runs

Total delay mass across the 482 1D runs is 117,100 ticks. Three runs — seeds 171, 294 and 361, with delays of 38,400, 32,000 and 10,450 ticks — account for 80,850 of it, or 69.0 %. Excluding them, the mean falls from 242.9 to 75.7, within 26 ticks of the floor. The reported standard deviation, 2,319, is roughly ten times its own mean.

The archived `paper_data.json` already contained `delay_C_minus_B/median = 50.0` in the same object as the mean. The correction was available in the paper's own data file from the day of posting.

3.6 Defect 5 — the denominator double-counts

The headline denominator of 1,845 pools three studies: 482 runs from the 1D main study, 1,165 from the ablation grid, and 198 from 2D.

The ablation is one-at-a-time: four parameters at three levels each, where one level of each sweep *is* the baseline. Reporting the shared baseline in all four rows is standard practice and is not the error. The error was summing all twelve rows to obtain a total. All four baseline rows are identical to the last decimal place — mean B tick 4416.0, mean C tick 4518.556701030928, mean final S 0.7521093934657955 — because they are the same 97 runs. Counted once per parameter sweep, those 97 runs enter the total four times instead of once, contributing $97 \times 3 = 291$ redundant counts.

Distinct runs: $1{,}165 - 291 = 874$ from the ablation, so $482 + 874 + 198 =$ **1,554**, not 1,845.

3.7 Defect 6 — the effect sizes, and the story told about them

This is the centrepiece, because it is the only defect where we did not merely make an error — we explained one.

Hedges' g is computed on `final_pop`, grouped by `final_mean_s`. But S multiplies uptake, efficiency, and maintenance directly (bonus coefficients 0.12, 0.70 and 0.35), so it causally determines final population. Grouping on S and measuring population largely measures our own chosen coupling constants.

Worse, the group sizes are degenerate. Recomputing from the archived summaries with the published thresholds:

Table 3. Effect-size group sizes, recomputed. Both published values reproduce exactly.

Geometry	Organized ($\bar{S} > 0.3$)	Disorganized ($\bar{S} < 0.1$)	g	Published
1D	482	18	9.4124	9.41
2D	198	**2**	3.2737	3.27

The 2D effect size rests on a comparison arm of *two runs*. No 2D run falls between the thresholds, so the split is degenerate rather than merely unbalanced.

The repository additionally reports a "combined" $g = 6.34$. That value is exactly $(9.41+3.27)/2$: the arithmetic mean of two effect sizes, which is not a valid pooling under any convention, and one of whose inputs has $n = 2$.

And in §4.3 of the original paper we explained the lower 2D value as reflecting "more spatial degrees of freedom." An artifact appeared, and we reached for physics to account for it rather than checking `len()`. We consider this the single most instructive sentence in the original manuscript.

3.8 Defect 7 — gate dependence is systematic

Auditing all nineteen reported quantities by tracing each computation path in source, nine depend directly on detector outputs and a tenth depends on one indirectly. Phase-D success rates (90 % versus 79 % across lipid supply) are withdrawn as rates, because they count a latched state whose latch time is set by the definedness floor. Only the *direction* of the difference survives, and it survives for a specific reason: two independent confounds — more post-transition time available at high lipid supply (+5,049 ticks), and greater generational depth (4.87 versus 4.74) — both predict the opposite sign to the one observed.

The audit also demoted two results we had ourselves placed in the surviving column, both from the section added in response to external review. All four results in that section are withdrawn or demoted; the section does not survive as physics. Two survive as arithmetic with their causal readings withdrawn, because their measurement windows were anchored on `phase_C_tick` — which Defect 3 established is not when the Map latched.

3.9 Defect 8 — a sign reversal

The reported correlation between division period and pattern stability was $\rho = +0.43$ pooled across lipid conditions. Stratified:

Table 4. A Simpson's paradox. Pooling reverses the sign.

Stratum	n	$\rho(T_{\mathrm{div}}, S)$
Lipid supply 0.008	100	-0.5970
Lipid supply 0.015	100	-0.7396
Lipid supply 0.025	100	-0.7537
Pooled (as published)	300	$+\mathbf{0.4278}$

Lipid supply moves both variables at once, so they covary positively across conditions; within a fixed supply, slower-dividing runs reach *lower* stability. The published value has the wrong sign. The structure is the classical one (Simpson, 1951; Blyth, 1972).

3.10 The corrected figure

Measured without the gate, the Clock precedes the Map in approximately 1 % of runs: $1/97 = 1.0\,\%$ by 1-tick resampling (100 seeds, of which 97 latched both predicates), 0.7 % by a threshold grid over archived timeseries ($n = 150$), and 0.0 % in 2D across all nine cells of a 3×3 threshold grid. Two independent methods, in close agreement.

We note that the 97 here and the 97 baseline runs of Defect 5 are an arithmetic coincidence of two unrelated experiments, not the same quantity.

4. Validating the corrected instrument

4.1 Why a null from an unvalidated instrument is worthless

At this point the audit had produced a series of null and negative results. None of them meant anything yet.

An instrument that cannot detect an ordering will report no ordering whether or not one exists. Reporting nulls from an uncalibrated detector is the same class of error as the original paper, pointed in the opposite direction — and we had committed it for several weeks without noticing.

4.2 Planting a known ordering

We planted an ordering in the *physics*, not the detector: pattern stability is held at zero for every cell until tick T, making spatial organization physically impossible before then. No involvement of the detector, the gate, or the Clock statistic; ground truth known exactly. A wall-clock plant was chosen deliberately, since gating the plant on division regularity would inherit the very floor under test.

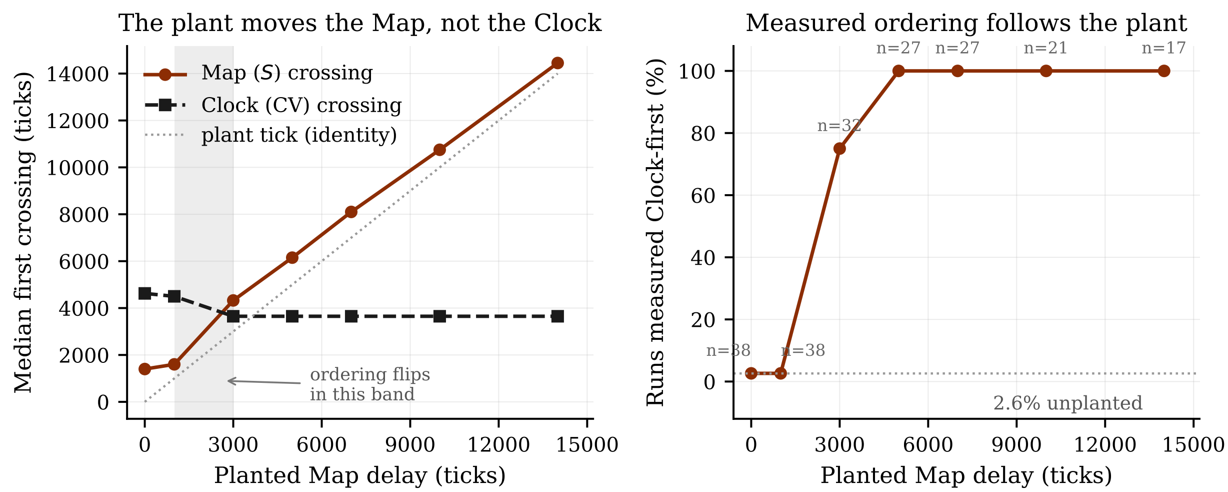

Figure 1. The positive control. **Left:** median first-crossing tick of each predicate against planted Map delay. The Map crossing tracks the plant (dotted identity line); the Clock crossing does not follow it, but does shift from 4,625 to 3,650 once the plant bites — the non-orthogonality noted below, and the reason ordering is judged within-run rather than against a fixed Clock time. **Right:** fraction of runs measured Clock-first, rising from 2.6 % to 100 % as the plant crosses the natural Clock time. Annotated n is the number of runs in which both predicates latched, which falls at large plants.

Table 5. Positive control. 280 runs, seven plant levels × 40 seeds. The measured ordering flips as the plant crosses the natural Clock time for this configuration (∼4,250 ticks; the archive median of 4,200 in Defect 3 is the same quantity at a different run length and seed count).

Plant tick	Median measured Map crossing	Clock-first	Map after plant
0 (none)	1,400	2.6 %	—
1,000	1,600	**2.6 %**	100 %
3,000	4,325	75.0 %	100 %
5,000	6,150	100 %	100 %
7,000	8,100	100 %	100 %
10,000	10,750	**100 %**	100 %
14,000	14,450	100 %	**100 %**

All three pre-registered criteria pass: recovery ($\rho = +0.9560$, required $> +0.8$; $p < 10^{-100}$, computed over 200 runs at the cell level and quoted here only as a bound, since at this effect size the exact value is an artifact of the sample size rather than evidence about the ordering); flip (2.6 % → 100 %, required < 20 % and > 80 %); and fidelity (100 %, required > 90 %).

4.3 Consequence

The corrected measurement detects a real ordering when one exists, and the crossover falls where it should. The ≈1 % figure is therefore a measurement of the model's dynamics, not a blind spot. It also disposes of the obvious objection — that the method is biased toward Map-first — since planting the ordering flips it to 100 % Clock-first without hesitation.

Three caveats belong in the record. Runs in which both predicates fired fall from 38/40 at no plant to 17/40 at the largest plant, because larger plants leave less run time for stability to recover; the recovery correlation therefore rests on 200 of 280 runs. The plant is not perfectly orthogonal: suppressing S also suppresses the S-linked metabolic bonuses, shifting division dynamics. Ordering is consequently judged within each run on that run's own two crossings, never against an assumed Clock time.

The third caveat bounds what the control licenses. It plants *Clock-before-Map* by delaying the Map, and demonstrates recovery in that direction only. We did not run the mirror experiment — an accelerated Clock, or a delayed Clock against an undisturbed Map — because the Clock statistic's definedness floor is precisely the quantity under investigation, so any plant acting on it would be confounded with the effect being measured. The control therefore establishes that the method is not blind to Clock-first orderings when they exist. It does not establish symmetric sensitivity, and the ≈1 % figure should be read as an upper bound on our ability to detect Clock-first rather than as a two-sided measurement.

5. Three attempts to rescue the hypothesis

The underlying physical claim — that temporal regularity enables spatial organization — had still never been tested, only mismeasured. We made three attempts.

5.1 Intervention

We imposed division timing externally, drawing intervals from $\Gamma(1/c^2, Tc^2)$, which has mean T and coefficient of variation c exactly ($c = 0$ deterministic, $c = 1$ Poissonian). Division regularity becomes a control variable rather than an estimate, and steady-state S is measured over the final 20 % of each run with no detector and no CV estimator anywhere in the causal path. Grid: nine CV levels × six periods × 25 seeds = 1,350 runs.

Only two of the six periods are physically valid: below the model's natural period under this forced-division configuration (∼3,170 ticks) forced division splits cells before they have grown, reaching 79.4 % premature splits at $T = 1600$ and 100 % at $T = 200$. (Elsewhere in this paper the natural period is given as ∼2,500 ticks. The two figures describe different experiments — the 1D main study at 80,000 ticks under natural division, and this forced-division grid — and are not in conflict. Every period quoted in this paper is configuration-specific and we label it as such from here on.) Within the valid window:

Table 6. Steady-state S against imposed division regularity, physical periods only ($n = 450$).

Imposed CV	0.0	0.05	0.10	0.15	0.20	0.30	0.50	0.80	1.0
Mean S	0.760	0.664	0.687	0.710	0.700	0.702	0.726	0.773	0.781

$\rho(\mathrm{CV}, S) = +0.2196$ ($p = 2.6 \times 10^{-6}$): stability *rises* with irregularity, opposite to prediction. A one-sided Mann–Whitney test of the hypothesis' own direction (regular > irregular) returns $p = 1.0$.

That monotone summary misdescribes the table it summarises. The response is not monotone but U-shaped: S falls from 0.760 at CV = 0 to a minimum of 0.664 at CV = 0.05, then rises to 0.781 at CV = 1.0. The perfectly regular condition is the second-*highest* in the row, not the lowest, so a rank correlation computed across the whole range averages a decrease and an increase and reports the net. We flag this rather than resolve it: the CV = 0 point is also the condition with 27.3 % extinction (§8), so the surviving runs there are a survivorship- filtered subpopulation and the point is not comparable to its neighbours on equal terms. The honest statement is that imposed regularity does not raise stability anywhere in this grid, and that the single number +0.2196 should not be read as a dose–response.

The effect survives three of four pre-registered confound checks, including a matched-interval control that excludes the obvious alternative explanation (the Gamma right tail): the association persists within interval-matched quintiles with only 19 % attenuation ($\rho = +0.169$ unconditional versus +0.137 within bins, $n = 139{,}715$). It fails the fourth: the fraction of barely-dividing cells rises from 44.0 % to 56.2 % across the sweep, against a pre-registered threshold of 10 percentage points. The hypothesis is therefore neither confirmed nor refuted by this experiment.

We note that we did not move the threshold. The primary confound was cleared cleanly and the miss was 2.3 points on a criterion we had set ourselves.

5.2 Reformulation

Perhaps the theory named the wrong variable. Variance per se has no obvious mechanism; short intervals do. If patterns need ∼1,350 ticks to consolidate, what should matter is the *minimum* uninterrupted interval, with regularity mattering only insofar as it produces short ones. This also predicts the flat result above, since raising Gamma CV adds both short and long intervals.

Pre-registered and run on natural division ($n = 153{,}133$ cell-observations): $\rho(\text{min interval}, S) = +0.3844$ pooled, which passes the criterion. Stratified by division count, it does not:

Table 7. The third Simpson's paradox. The pooled association vanishes or inverts within strata.

Division intervals	n	ρ(min interval, S)
1–2	28,266	+0.0918
3–4	25,745	−0.1525
5–8	50,375	−0.0495
9+	48,747	−0.0840
Pooled	153,133	**+0.3844**

Cells that barely divided have both an upward-biased minimum and an undisturbed high stability, so pooling manufactures the correlation. The pre-registered disqualifier fires, and the secondary arm (forced division) gives $\rho = -0.1484$ pooled, *strengthening* to -0.3463 under stratification — the mirror image, and therefore not an artifact. Both arms agree that longer minimum intervals do not produce more stable patterns.

5.3 Redesign, and what it revealed

The remaining possibility was that the instruments themselves were mismatched. We therefore attempted to replace the Clock metric with one having no definedness floor: the normalised autocorrelation of a cell's own radius trajectory. A regularly dividing cell traces a sawtooth that autocorrelates strongly at its period; an irregular one weakly. It requires no divisions and is defined as soon as its buffer fills.

Any such metric must first pass a validity check: clock_r must correlate *negatively* with CV across runs, since a more regular divider should show both higher autocorrelation and lower interval variance. A metric failing this is not measuring regularity, whatever its definedness properties.

The first parameterisation failed it: with a 400-tick buffer searching lags of 30–200 ticks, $\rho(\text{clock}_r, \text{CV}) = -0.190$, $p = 0.241$ over 40 seeds. The division period in this configuration is $\sim$2,550 ticks and the lag window could not reach it.

Widening the buffer to 6,000 ticks — 2.4 division periods — does not fix it. Over 40 seeds that parameterisation gives $\rho = +0.2162$, $p = 0.186$: still the wrong sign, still insignificant. An earlier draft of this paper reported $\rho = -0.2364$ for this arm and treated it as validation. That figure is the first ten seeds, it was never significant ($p = 0.511$), and it decays to -0.0822 ($p = 0.614$) at $n = 40$. Quoting a correlation without its p-value to establish that an instrument works is the error this paper is about, committed by us, in the section documenting it. It is the sixth entry in §6.3.

We therefore swept the buffer to ask whether *any* span yields a valid metric, holding the sample count fixed at 120 so that only the span varies and reporting all four arms rather than stopping at the first that passed.

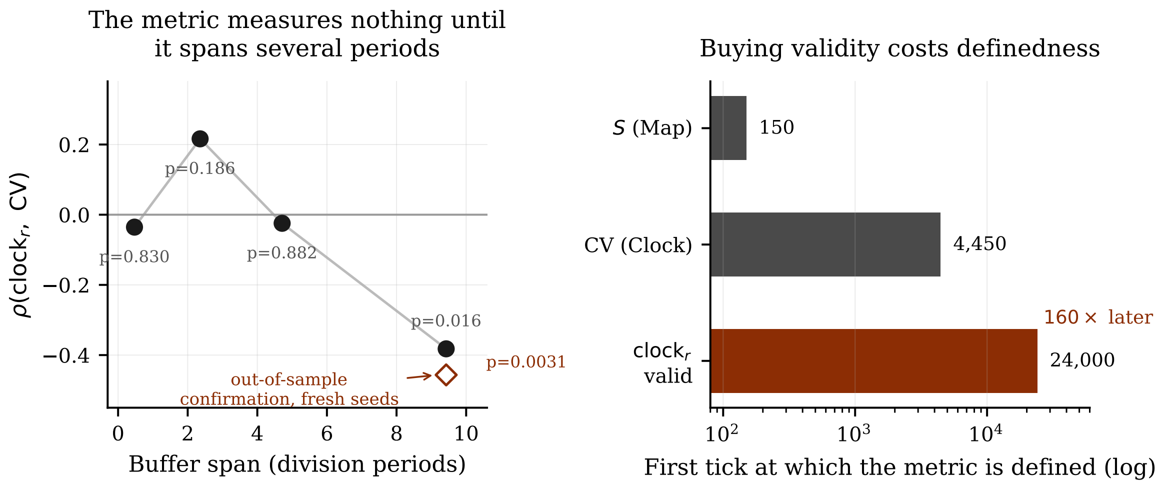

Figure 2. Why the redesign could not work. **Left:** validity of clock_r against buffer span. The metric does not track division regularity until its history spans ∼9 periods; the 2.4-period arm has the wrong sign. Filled points are the selection sweep (black: does not clear the four-arm Bonferroni threshold); the open diamond is the widest arm re-run on 40 fresh seeds. **Right:** first tick at which each metric is defined, log scale. Because definedness equals buffer length identically, the span that buys validity is also the delay it costs.

Table 8. Buffer sweep, 40 seeds per arm, 40,000-tick runs. Validity requires $\rho(\text{clock}_r, \text{CV}) < 0$ and significant. Definedness equals buffer length in every arm.

Buffer	Periods	Defined at	ρ	p	Valid
1,200	0.47	1,200	−0.0354	0.831	no
6,000	2.36	6,000	+0.2162	0.186	no
12,000	4.71	12,000	−0.0245	0.882	no
24,000	9.43	24,000	**−0.3822**	0.016	yes

Two features carry the argument. **Definedness equals buffer length exactly**, which is an identity rather than a result: clock_r exists the moment its buffer fills. Whatever span validity demands is therefore also the earliest the metric can exist. And **the approach is not monotone** — the 2.36-period arm has the wrong sign and the 4.71-period arm is flat — so we claim no smooth relationship between span and validity, only that the intermediate arms are uninformative at $n = 40$.

The widest arm's $p = 0.016$ does not clear a Bonferroni threshold of $0.05/4 = 0.0125$. Since this is the paper's central claim, we re-ran that arm on 40 seeds it had never been evaluated on: $\rho = -0.4565$, $p = 0.0031$, with clock_r defined at tick 24,000 in 40 of 40 runs. That is a single pre-specified test of a directional prediction and needs no correction.

Table 9. Definedness of each metric, medians. The replacement metric, in the only parameterisation shown to measure anything, is defined 160× later than the metric whose floor it was built to remove.

Metric	Defined at tick	Ratio to S
S (Map) — spatial correlation	150	—
CV (original Clock)	4,450	30×
clock_r, *valid* parameterisation	**24,000**	**160×**

The pre-registered symmetry criterion required agreement within 20 %. The observed ratio is 160×. The redesign did not merely fail to remove the definedness floor it was built to remove; made valid, it raised that floor by a further factor of five. Per the pre-registered decision rule the downstream tests were not computed.

One property of the valid measurement must be stated rather than left to be discovered. A 24,000-tick buffer is filled only by cells that have lived 24,000 ticks, and daughters begin with an empty buffer while mothers retain theirs, so clock_r is an average over an age-selected subpopulation. This does not rescue the metric. A cell too young to have a full buffer is exactly a cell whose period has not yet been observed; the selection restates the constraint rather than concealing it.

5.4 The finding

The redesign could not have worked, for a reason that has nothing to do with implementation.

To measure how regular a periodic process is, one must observe several of its periods. This is a time–frequency constraint of the same family as the classical bound on simultaneous time and frequency resolution (Gabor, 1946), and it is not specific to either estimator we tried. The CV of division intervals needs four divisions. A radius autocorrelation, measured here, needed ∼9 periods before it tracked regularity at all — two were not enough.

We state the scope of this claim carefully, because the obvious objection is a good one. It applies to *per-lineage* regularity statistics: any quantity asking how evenly spaced one cell's events are must wait for several of that cell's events. It does *not* follow that no Clock-like quantity can ever be defined early. A *cross-sectional* measure — the dispersion of division phase across a population at one instant — is defined after roughly one period rather than several, and we have not tested one. Our claim is therefore bounded: the floor for per-lineage regularity is of order several division periods, which for this model is ∼24,000 ticks against S at ∼150; whether a population-level phase statistic narrows that gap is open, and we regard it as the most promising line for anyone wishing to salvage a timing-based approach.

Meanwhile S measures spatial correlation of a reaction–diffusion field whose intrinsic correlation time is ∼40 ticks, and is defined by ∼150. The asymmetry is therefore set by the ratio of the two phenomena's intrinsic timescales — a slow periodic process against a fast field process — and not by any choice we made.

Defect 3 is consequently only half an error. The specific four-division floor is ours and could be loosened. The existence of a floor of order several division periods is physical and unavoidable. No simulation redesign removes it.

5.5 Therefore the question is malformed as posed

Asking "does the Clock precede the Map?" *as a comparison of first-crossing times* compares two quantities that become knowable at different times by their nature. The comparison measures the timescale ratio, not the dynamics.

This accounts, in one stroke, for why every approach in this audit failed in the same way. The gate, the definedness floor, the imposed-CV grid, the interval-floor reformulation — all were attempts to time-order two phenomena that cannot be timed against one another.

5.6 The constructive replacement

Ordering claims of this kind must be tested by *intervention* rather than by timing:

> *Does perturbing division regularity change spatial organization?*

That question is causal, has no definedness problem, and is exactly what §5 asked by imposing CV as a control variable. Its answer was no support for the hypothesis. The theory has therefore been tested by the method that can test it, and not by the method that never could have been.

A laboratory escapes the constraint entirely. A microscope tracking individual lineages records *events* rather than estimating summary statistics, so both quantities are observable from $t = 0$ without waiting for anything to become computable. We give a protocol in the supplement.

6. Why review did not catch this

6.1 What review did catch

The original paper went through three rounds of adversarial review by frontier language models, and the reviews were not poor. They identified genuine overreach in the conclusions, missing engagement with the assembly-theory literature (Marshall et al., 2021), an unasked question about whether spatial organization might destabilise the division rhythm that enabled it, and an implicit assumption about the ratio of division period to pattern-consolidation time. Each prompted real additions. One prompted an entire new section.

6.2 What it could not catch

None of the three found any of the eight defects.

The reason is structural rather than a failure of diligence. Conceptual review reads the *argument*: is the logic sound, are the claims proportionate, is the literature engaged, does the conclusion follow? Every defect documented here lived in the *instrument*. Finding them required pointing code at code and asking whether the detector could have emitted a different answer — an activity nobody performed, human or machine, for four months.

There is a sharper way to state it. The reviews improved the paper's argument for a result that did not exist. The most thorough round added a section which this audit has now withdrawn in its entirety.

6.3 The auditors repeated the failure

We consider this the most instructive part of the account, and we state it in full.

During the audit we published six claims to the public repository that we subsequently withdrew. They fall into three classes.

Wrong reference point (three occurrences). In each, we confirmed that a number reproduced without checking what it was a number *of*.

1. We described a 2D back-reaction result as "the most interesting surviving number in the project." It does not replicate ($p = 0.427$).
2. We corrected a published figure of "roughly 4 %" to 22.8 % — correct arithmetic applied to a result that should have been withdrawn outright.
3. We placed two results in the surviving column whose measurement windows were anchored on `phase_C_tick`, a detector output. Their arithmetic is exact; their causal readings are not licensed.

Wrong scope (two occurrences). In each, a correct number was generalised past its domain.

4. We claimed that "∼50 % of cells in every condition barely divide, so division-perturbation effects cannot be isolated in this model at all," and described it as a property of the simulation class. It is a property of one forced-division design (44–56 %). Under natural division the figure is 11–13 %.
5. We proposed a timescale-separation mechanism from a comparison that contrasted an unphysical condition against a physical one.

Unvalidated instrument (one occurrence), which is the original paper's own error and the one we would least have predicted committing.

6. In §5.3 we reported $\rho(\text{clock}_r, \text{CV}) = -0.2364$ as establishing that the redesigned Clock metric measured regularity, and built the section's conclusion on it. We did not quote its p-value, which was 0.511. The figure was ten seeds; at forty it is -0.0822 ($p = 0.614$), and the

parameterisation fails the validity check outright. We had written a section arguing that nulls from unvalidated instruments are worthless (§4) and then, four subsections later, validated an instrument by eye from an insignificant correlation. The error was caught only when the experiment was re-run with saved outputs, because the original had been executed inline and its numbers existed nowhere but in the prose.

A seventh case deserves separate mention, being a near-miss rather than a withdrawal: a pre-registered decision rule returned the correct verdict for the wrong reason. Applied literally it would have preserved claim (4). We report it because pre-registration is frequently offered as a remedy for exactly this failure mode, and here it was not sufficient on its own.

6.4 Implication

Adversarial review of claims is not a substitute for mechanical audit of instruments. And the auditor requires the same mechanical checks as the author, because the auditor is subject to the same failure — as demonstrated six times above, by people actively looking for it, in a document whose subject is that failure.

Entry (6) additionally identifies the mechanism. Every one of the eight original defects, and five of our six, survived because a number was inspected without its provenance. The countermeasure that actually worked was not vigilance but a file: the error was found when the experiment was re-run as a script that wrote its outputs to disk, at which point the correlation could be recomputed at a larger n by anyone, including us. Numbers that exist only in prose cannot be audited, and we had several.

7. An automated checklist

Every defect documented here is mechanically detectable. We release `detector_audit.py`, which encodes eight checks:

- **C1** any predicate whose latching is conditioned on another predicate's state
- **C2** any metric with a definedness floor
- **C3** any reported delay whose median equals the sampling interval
- **C4** any summary reported mean-only where $|\mathrm{SD}| > |\mathrm{mean}|$
- **C5** any effect-size dependent variable causally downstream of its grouping variable
- **C6** any effect size with a group of $n < 10$, or grouping thresholds leaving an unpopulated gap
- **C7** any reported quantity that does not declare its reference point, population and temporal window

- **C8** any pooled statistic whose sign or magnitude does not survive stratification by an entanglement variable

C1–C6 encode defects found by review. C7 and C8 encode the failure modes of the *audit*: C7 catches the wrong-reference-point class, and flags as a finding any quantity anchored on a detector output. C8 was added after three Simpson's paradoxes passed C1–C7 untouched.

Run against this repository the tool reports 30 findings, 15 critical, and recovers all eight defects from source and archived data with no audit knowledge encoded. It independently locates the $n = 2$ comparison group that took three rounds of expert review and four months to notice by eye, and it flags precisely the two results the authors wrongly retained.

Three checks are regular-expression based and tuned to this codebase's idioms; the tool is a floor, not a ceiling.

7.1 Why C8 generalises

In a model of a dividing population, nearly every quantity is entangled with how many times a cell has divided: stability is disrupted by division, the CV requires divisions to exist, minimum interval is taken over a division-count-dependent sample, and final population *is* division count. Division count is a hidden common cause behind almost every pair of variables, so pooling across it produces sign reversals as a matter of course.

Three occurred here. We expect the same structure in any agent-based model of a reproducing population, and consider this the most transferable technical result in the paper.

8. What survives

The following results are unaffected by the audit and are reported with the scrutiny applied to everything else:

- **The generalization argument.** Pattern stability measures persistent spatiotemporal correlation rather than classical Turing bifurcation (Turing, 1952; Kondo and Miura, 2010), so any ordering principle expressed in terms of it applies to phase-separated condensates and lipid microdomains, not only to Gray–Scott chemistry. Conceptual; untouched.
- **The timescale anchor.** Fixed *a priori* from lipid lateral diffusion ($D \approx 5\,\mu\text{m}^2/\text{s}$ on a $1.5\,\mu$m vesicle), one tick ≈ 0.45 s, giving division periods of 15.8–38.3 minutes across the lipid-supply range. This brackets *E. coli* doubling time (Wang et al., 2010) and overlaps published fatty-acid vesicle division times (Zhu and Szostak, 2009). The anchor was set without reference to the division data, so the agreement is a check rather than a fit.
- **One lipid-supply result, measured twice.** Higher lipid supply is associated with lower pattern stability, seen as final S falling $0.823 \to 0.644$, and restated in thresholded form by

Phase-D attainment (direction only). These are not two findings and certainly not three; the third measure was $\rho = +0.43$, which is withdrawn.

The mechanism is not identified. An earlier draft of this section wrote the result as "higher supply → faster division → lower stability," asserting mediation through division rate. Our own Table 4 refutes it: within every lipid stratum, slower division goes with *lower* stability ($\rho = -0.60$ to -0.75), which is the opposite of what that chain predicts. Attributing a between-condition association to a plausible intermediate variable, in the presence of stratified data contradicting it, is Defect 6 committed again — and it survived into a draft of the section titled "What survives." What is supported is the association and its direction, nothing about the path.

- **Extinction under forced fast division.** 32.4 % population extinction at the shortest imposed period, 8.4 % at the next, zero at periods ≥ 1600.
- **Extinction under enforced synchrony.** 27.3 % extinction at imposed CV = 0 versus 1.3–4.0 % at CV ≥ 0.1. Lockstep division depletes shared resource simultaneously; heterogeneity leaves some lineages off-phase when the trough arrives. This is counterintuitive and, to our knowledge, untested experimentally.

What is not claimed. We do not claim the Clock precedes the Map. We do not claim the reverse; the apparent Map-first signal is partly the same definedness floor. We do not claim the physical hypothesis is false — it remains untested, and the direct test could neither confirm nor refute it. We claim only that the published result was an artifact, that the question cannot be settled by the method used, and that the surviving measurements are the ones listed above.

9. Discussion

9.1 Perfect separation as a warning

1,845 out of 1,845 was treated as the strongest feature of the original result. It should have been treated as the first thing to explain. Biological and physical systems with genuine variability rarely produce perfect separation across geometries, parameter regimes and noise levels; when they appear to, the most economical hypothesis is that the measurement cannot express the alternative.

We suggest a working rule: a result of exactly 100 % should trigger an audit of the detector before it triggers a paper. This is a detector-side counterpart to the base-rate arguments about why published findings so often fail to hold (Ioannidis, 2005; Open Science Collaboration, 2015; Baker, 2016): those concern the statistics of claims, whereas the failure here is upstream of any statistic.

9.2 Pre-register the instrument, not only the analysis

Pre-registration conventionally covers hypotheses, thresholds and analysis plans (Nosek et al., 2018; Munafò et al., 2017), and is the standard remedy proposed for undisclosed analytic flexibility (Simmons et al., 2011; Gelman and Loken, 2014). Every defect here would have survived such a pre-registration untouched, because all of them lived upstream of the analysis, in the measurement. Our own experience is a caution: §6.3 records a pre-registered decision rule that returned the right verdict for the wrong reason.

What was needed was a pre-registered *instrument validation*: a demonstration, before any result is reported, that the detector can recover a planted effect of the kind under study. We performed that validation only after publishing several nulls, and we were fortunate it passed. Had it failed, conclusions already public would have required retraction.

9.3 On publishing the correction with the data

The complete audit — pre-registrations, analysis scripts, raw outputs, negative results and the authors' own withdrawn claims — is public in the repository. This was not a costless choice, but the alternative was worse: the original claim was live and citable, and its evidence was already public. Anyone who re-ran the archived data could have found what we found.

It is worth stating what made this possible. Of the three companion papers in this series, two cannot be reproduced from surviving materials: no source code exists for either, and one results file lacks the column needed to trace its headline statistic. Those results are permanently unverifiable. This one was verifiable, and therefore correctable, only because its code was published.

9.4 Cost

The audit consumed roughly four months of intermittent effort. The tool that recovers all eight defects runs in under a second. Most of the four months was spent discovering *which* checks to write; running them is free.

10. Conclusion

We reported a simulation result that was reproducible, robust across parameters, replicated across geometries, statistically overwhelming, and wrong. The detector could not emit a violation, so the perfect score was the only reachable output; the reported delay was the sampling interval; and the reported transition time was the moment a statistic became computable.

Corrected, the ordering holds in approximately 1 % of runs — a figure we validated by planting a known ordering and recovering it. Three attempts to rescue the underlying hypothesis failed, and the third revealed why: the regularity of a slow periodic process and the correlation of a fast spatial field become measurable at different times by their nature, so comparing their first-crossing times measures a timescale ratio rather than a dynamics. The question is malformed as posed and must be asked by intervention instead.

The most useful output is not a corrected number. It is the observation that eight defects survived three rounds of competent adversarial review because review reads arguments and none of these defects was in the argument — together with a checklist that finds all eight in under a second, including the two the authors themselves got wrong while writing this paper.

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Data and code availability

All simulation code, raw outputs, analysis scripts, pre-registrations and audit reports are public at `https://github.com/AVADSA25/genesis-engine`, including every negative result and every claim the authors withdrew during the audit. The checklist runner is at `tools/detector_audit.py`; the laboratory protocol at `paper/v6_supplement/LAB_PROTOCOL.md`.

The repository was public throughout the period in which the withdrawn claim was live, and the archived `paper_data.json` contained `median = 50.0` beside the reported mean from the day of posting.

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